PYTHON ACTVITY BASED LEARNING

Problem-2:

Nested Loop Examination Seating Generator: Generate a seating arrangement with row and column identifiers, reserved positions, alternate seating restrictions, and fixed classroom capacities.

Objective: To generate an examination seating arrangement using nested loops by assigning row and column identifiers, identifying reserved seats, enforcing alternate seating restrictions, and maintaining fixed classroom capacity.

Algorithm

1. Start.
2. Read the number of rows and columns from the user.
3. Calculate classroom capacity as rows × columns.
4. Read the number of reserved seats.
5. Input the row and column positions of each reserved seat and validate them.
6. Initialize student_no = 1, occupied = 0, reserved_count = 0, and empty = 0.
7. Use nested loops to traverse each row and column.
8. For each seat:
    - If the seat is reserved, mark it as RESERVED and increment reserved_count.
    - Else if the column number is even, mark it as EMPTY and increment empty.
    - Otherwise, assign a student ID, increment student_no, and increase occupied.
9. Display each seat with its row and column identifiers.
10. Display the total capacity, students seated, reserved seats, and empty seats.
11. Stop.

Pseudo code:

```
BEGIN

  INPUT rows, cols
  capacity = rows * cols
```

```
INPUT n
reserved = empty list

FOR i = 1 TO n DO
    INPUT r, c

    IF r >= 1 AND r <= rows AND
       c >= 1 AND c <= cols THEN
        ADD (r, c) TO reserved
    ELSE
        DISPLAY "Invalid seat position"
    END IF
END FOR

student_no = 1
occupied = 0
reserved_count = 0
empty = 0

FOR r = 1 TO rows DO
    FOR c = 1 TO cols DO

        IF (r, c) IS IN reserved THEN
            seat = "RESERVED"
            reserved_count = reserved_count + 1

        ELSE IF c MOD 2 = 0 THEN
            seat = "EMPTY"
```

```
            empty = empty + 1

        ELSE
            seat = "S" + student_no
            student_no = student_no + 1
            occupied = occupied + 1
        END IF

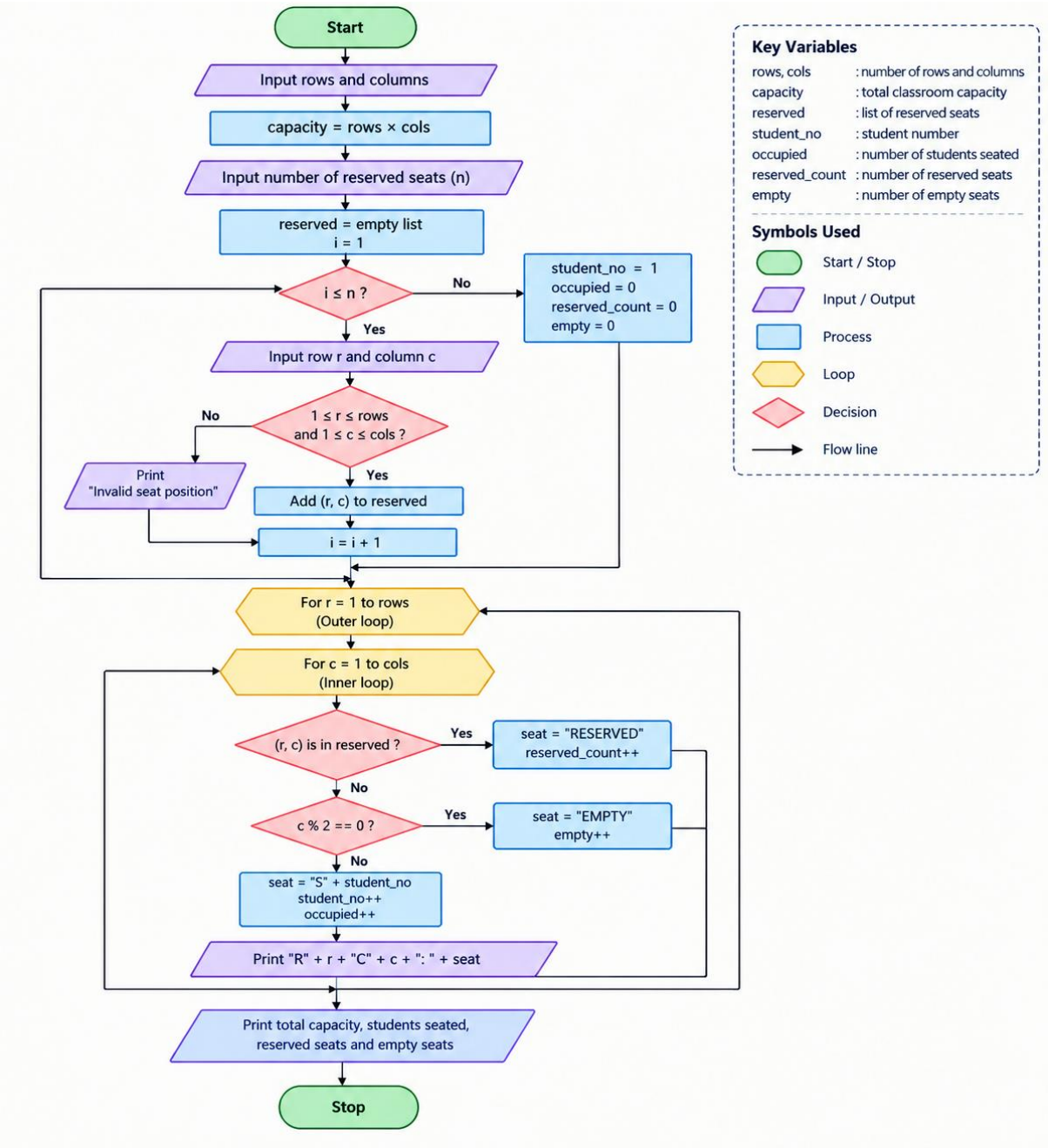
        DISPLAY row, column, seat

    END FOR
    DISPLAY new line
END FOR

DISPLAY "Total capacity:", capacity
DISPLAY "Students seated:", occupied
DISPLAY "Reserved seats:", reserved_count
DISPLAY "Empty seats:", empty

END
```

Flow chart:

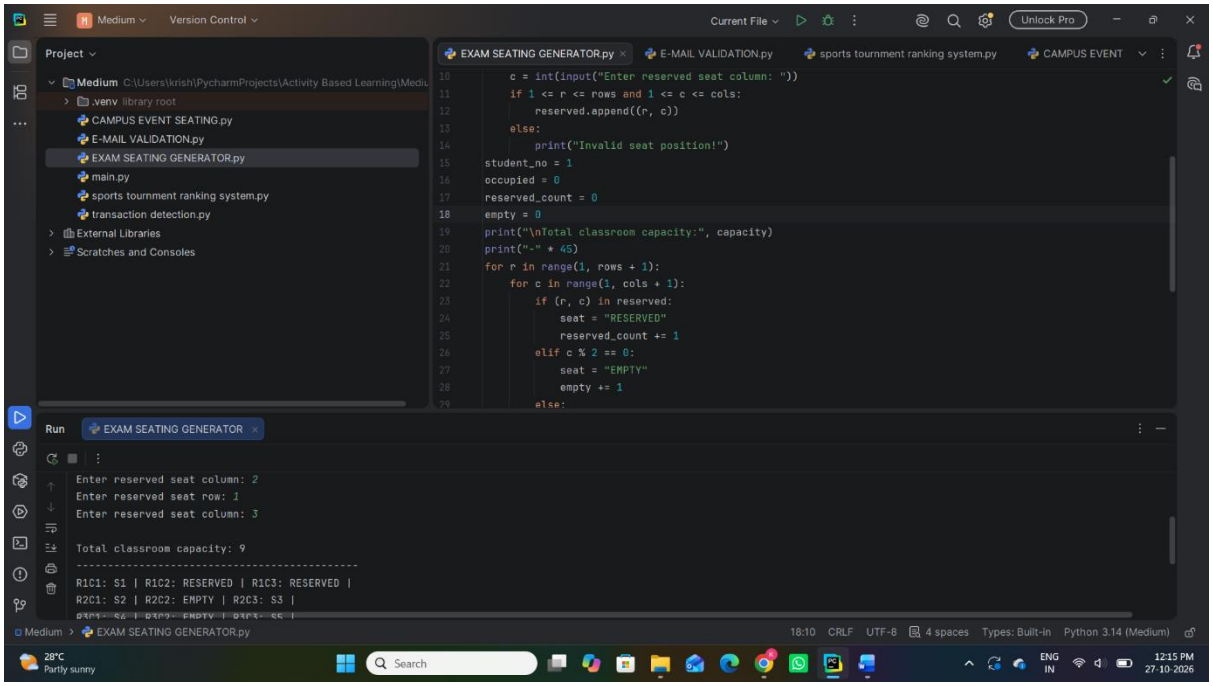


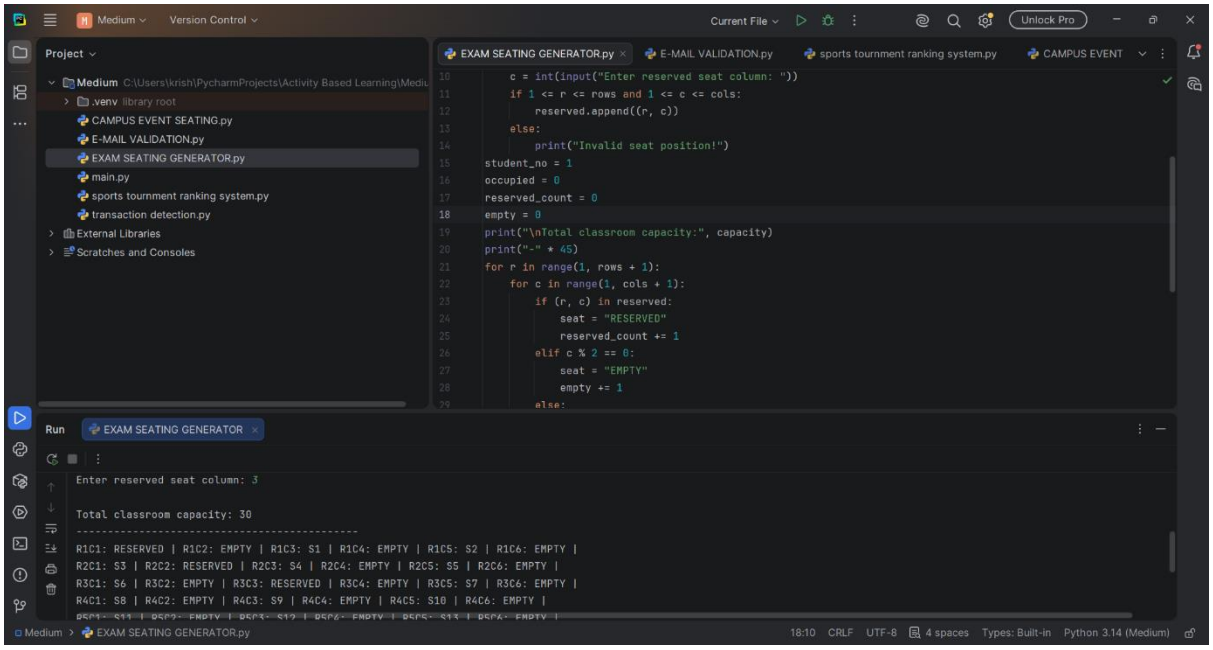
Python program:

```
print("EXAMINATION SEATING ARRANGEMENT")
print("-" * 45)
rows = int(input("Enter number of rows: "))
cols = int(input("Enter number of columns: "))
capacity = rows * cols
reserved = []
n = int(input("Enter number of reserved seats: "))
for i in range(n):
    r = int(input("Enter reserved seat row: "))
    c = int(input("Enter reserved seat column: "))
```

```
    if 1 <= r <= rows and 1 <= c <= cols:
        reserved.append((r, c))
    else:
        print("Invalid seat position!")
student_no = 1
occupied = 0
reserved_count = 0
empty = 0
print("\nTotal classroom capacity:", capacity)
print("-" * 45)
for r in range(1, rows + 1):
    for c in range(1, cols + 1):
        if (r, c) in reserved:
            seat = "RESERVED"
            reserved_count += 1
        elif c % 2 == 0:
            seat = "EMPTY"
            empty += 1
        else:
            seat = "S" + str(student_no)
            student_no += 1
            occupied += 1
        print("R" + str(r) + "C" + str(c) + ": "
              + seat, end=" | ")
    print()
print("-" * 45)
print("Students seated:", occupied)
print("Reserved seats:", reserved_count)
print("Empty seats:", empty)
print("Total classroom capacity:", capacity)
```

Output Screen shots:


Medium
Version Control
Current File
Unlock Pro
Project
Medium C:\Users\krish\PycharmProjects\Activity Based Learning\Mediu
.venv library root
CAMPUS EVENT SEATING.py
E-MAIL VALIDATION.py
EXAM SEATING GENERATOR.py
main.py
sports tournment ranking system.py
transaction detection.py
External Libraries
Scratches and Consoles
EXAM SEATING GENERATOR.py
E-MAIL VALIDATION.py
sports tournment ranking system.py
CAMPUS EVENT
c = int(input("Enter reserved seat column: "))
if 1 <= r <= rows and 1 <= c <= cols:
reserved.append((r, c))
else:
print("Invalid seat position!")
student_no = 1
occupied = 0
reserved_count = 0
empty = 0
print("\nTotal classroom capacity:", capacity)
print("-" * 45)
for r in range(1, rows + 1):
for c in range(1, cols + 1):
if (r, c) in reserved:
seat = "RESERVED"
reserved_count += 1
elif c % 2 == 0:
seat = "EMPTY"
empty += 1
else:
Run
EXAM SEATING GENERATOR
Enter reserved seat column: 2
Enter reserved seat row: 1
Enter reserved seat column: 3
Total classroom capacity: 9
R1C1: S1 | R1C2: RESERVED | R1C3: RESERVED |
R2C1: S2 | R2C2: EMPTY | R2C3: S3 |
Medium
EXAM SEATING GENERATOR.py
18:10
CRLF
UTF-8
4 spaces
Types: Built-in
Python 3.14 (Medium)
28°C
Partly sunny
Search
ENG
IN
12:15 PM
27-10-2026


Medium
Version Control
Current File
Unlock Pro
Project
Medium C:\Users\krish\PycharmProjects\Activity Based Learning\Mediu
.venv library root
CAMPUS EVENT SEATING.py
E-MAIL VALIDATION.py
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External Libraries
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EXAM SEATING GENERATOR.py
E-MAIL VALIDATION.py
sports tournment ranking system.py
CAMPUS EVENT
c = int(input("Enter reserved seat column: "))
if 1 <= r <= rows and 1 <= c <= cols:
reserved.append((r, c))
else:
print("Invalid seat position!")
student_no = 1
occupied = 0
reserved_count = 0
empty = 0
print("\nTotal classroom capacity:", capacity)
print("-" * 45)
for r in range(1, rows + 1):
for c in range(1, cols + 1):
if (r, c) in reserved:
seat = "RESERVED"
reserved_count += 1
elif c % 2 == 0:
seat = "EMPTY"
empty += 1
else:
Run
EXAM SEATING GENERATOR
Enter reserved seat column: 3
Total classroom capacity: 30
R1C1: RESERVED | R1C2: EMPTY | R1C3: S1 | R1C4: EMPTY | R1C5: S2 | R1C6: EMPTY |
R2C1: S3 | R2C2: RESERVED | R2C3: S4 | R2C4: EMPTY | R2C5: S5 | R2C6: EMPTY |
R3C1: S6 | R3C2: EMPTY | R3C3: RESERVED | R3C4: EMPTY | R3C5: S7 | R3C6: EMPTY |
R4C1: S8 | R4C2: EMPTY | R4C3: S9 | R4C4: EMPTY | R4C5: S10 | R4C6: EMPTY |
Medium
EXAM SEATING GENERATOR.py
18:10
CRLF
UTF-8
4 spaces
Types: Built-in
Python 3.14 (Medium)